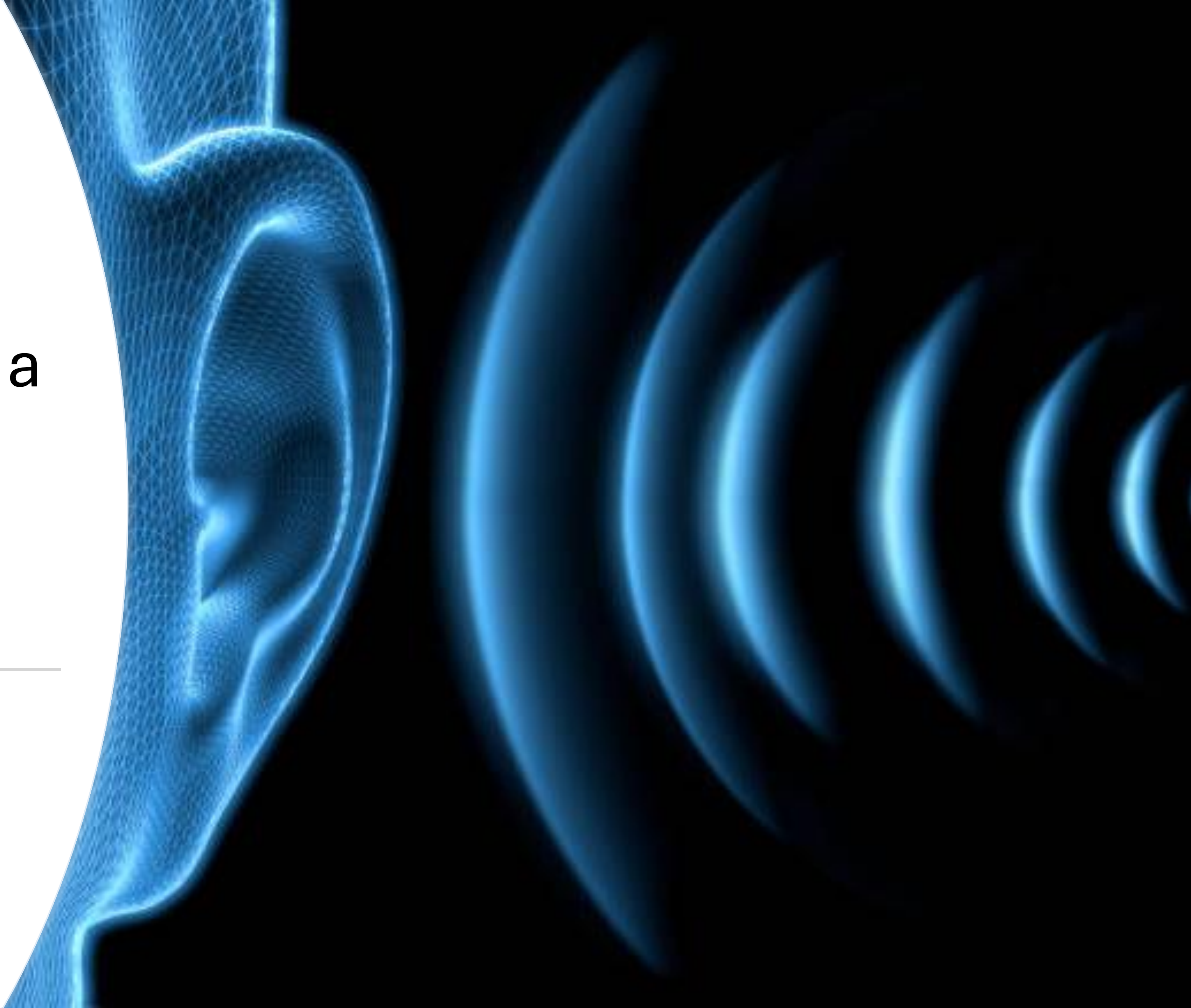
Design and Implementation of a Highly Directional Loudspeaker System



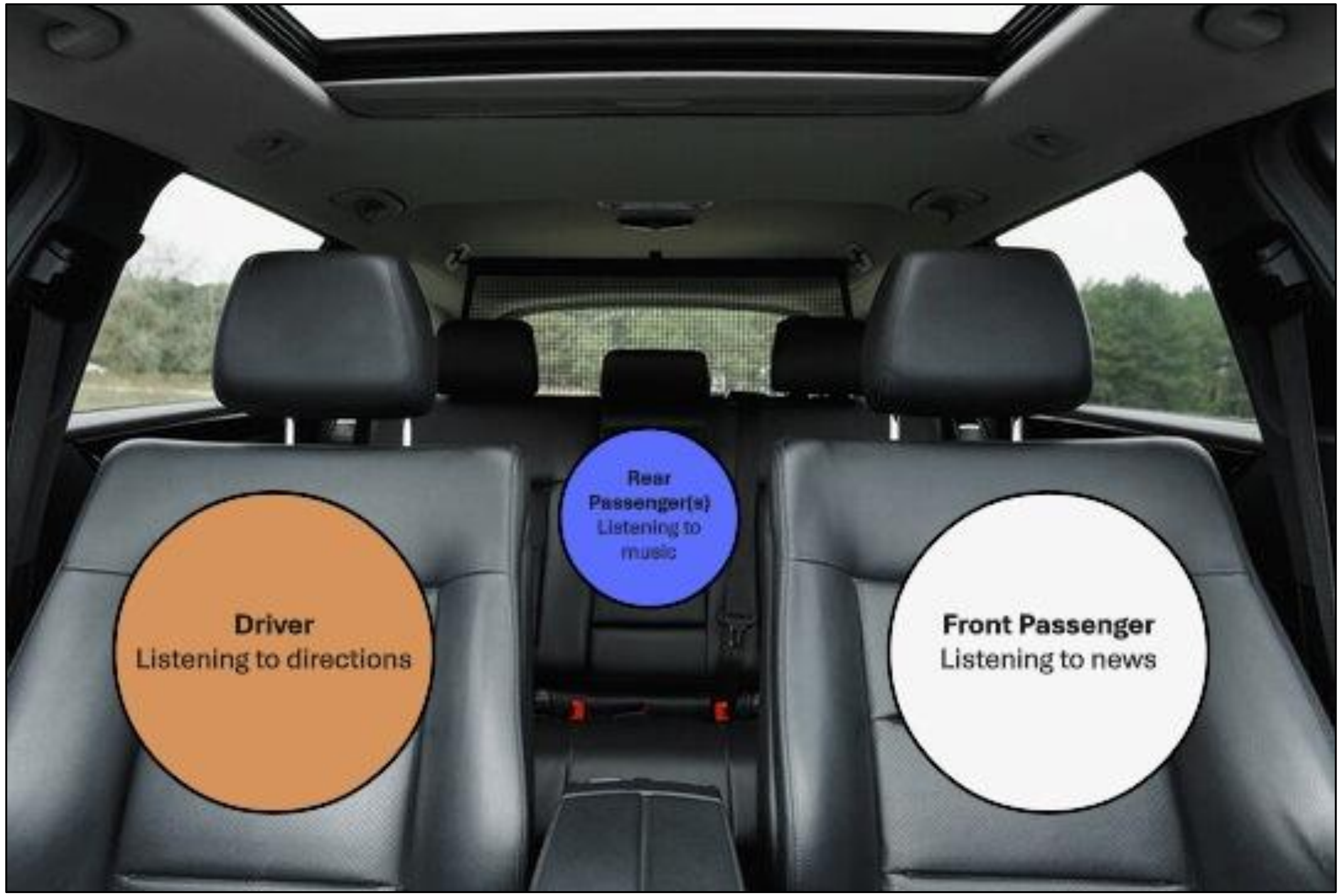
Midway Progress Presentation

Daniel Scully – 21431856

Supervisor: Prof. Martin Glavin

Project Motivation

- Different people may want to listen to different audio at the same time
- In a car: driver may need navigation instructions while passengers listen to music, etc.
- Watching the same film: each viewer could hear in a different language
- Public spaces: targeted announcements or information without disturbing others
- Directional speaker arrays can enable independent audio zones



Project Aim & Objectives

Aim:

To design and implement a highly directional loudspeaker system using beamforming principles

Create multiple audio zones while minimising sound leakage and interference

Objectives:

System Design: Create a theoretical and simulated loudspeaker array to analyse beamforming and sound direction control

Prototype Development: Build a prototype with hardware and software capable of controlling signal delay, phase, and amplitude

Testing & Evaluation: Measure directional sound performance, assess audio leakage and refine the design based on results

Documentation: Record the full process in project reports and deliverables

Project Scope

Work Package	Key Tasks
WP1: Background Research & Foundation Techniques	Background research, foundation techniques, component research
WP2: System Design & Simulation	Array design, beamforming simulation, parameter optimisation
WP3: Prototype Development & Implementation	Hardware assembly, signal control, multi-zone beam steering
WP4: Testing & Evaluation	Acoustic measurement, performance evaluation, system optimisation
WP5: Documentation & Deliverables	Project documentation, presentation materials

WP #	Task #	Sept	Oct	Nov	Dec	Jan	Feb	Mar	April
WP1	T1	X	XXX						
	T2		XX	XX					
	T3			XX					
WP2	T1		XX	XXXX					
	T2		XX	XXXX					
	T3		XX	XXXX	X				
WP3	T1					XX			
	T2					X	XXX		
	T3					X	XXX		
WP4	T1						XX	XX	
	T2						XX	XX	
	T3						XX	XX	
WP5	T1							XX	XX
	T2							XX	XX

Technical Background

•**Directional Audio / Beamforming**

•Focuses sound in a particular direction using multiple speakers

•Reduces sound spill to other areas

•Achieved by controlling signal timing, phase and amplitude

•**Speaker Arrays**

•Multiple speakers arranged in a pattern to control sound direction

•Spacing and number of speakers affect beam width and coverage

•**Constructive & Destructive Interference**

•Signals combine to strengthen sound in target zones (constructive)

•Signals cancel out to reduce sound in other areas (destructive)

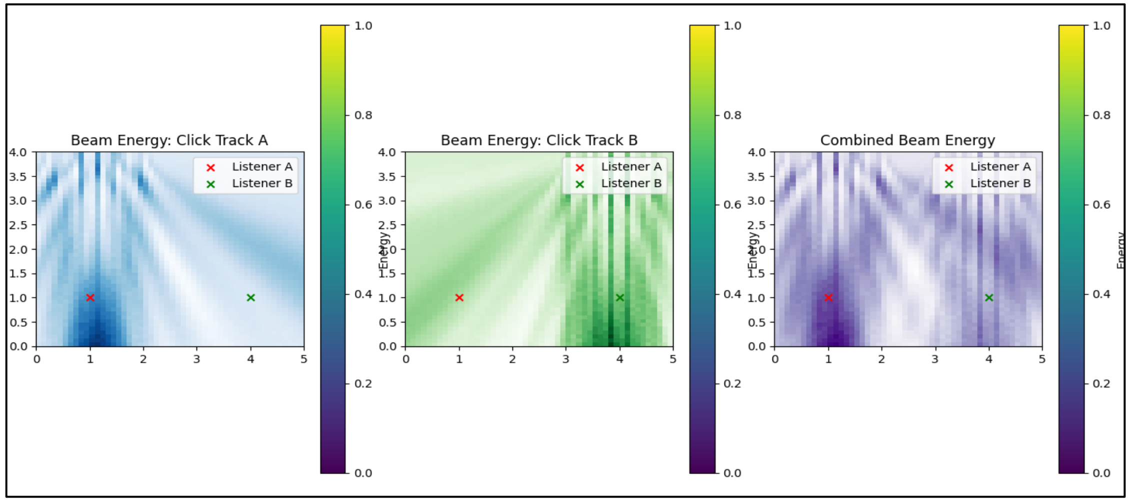
•**Multi-Zone Audio**

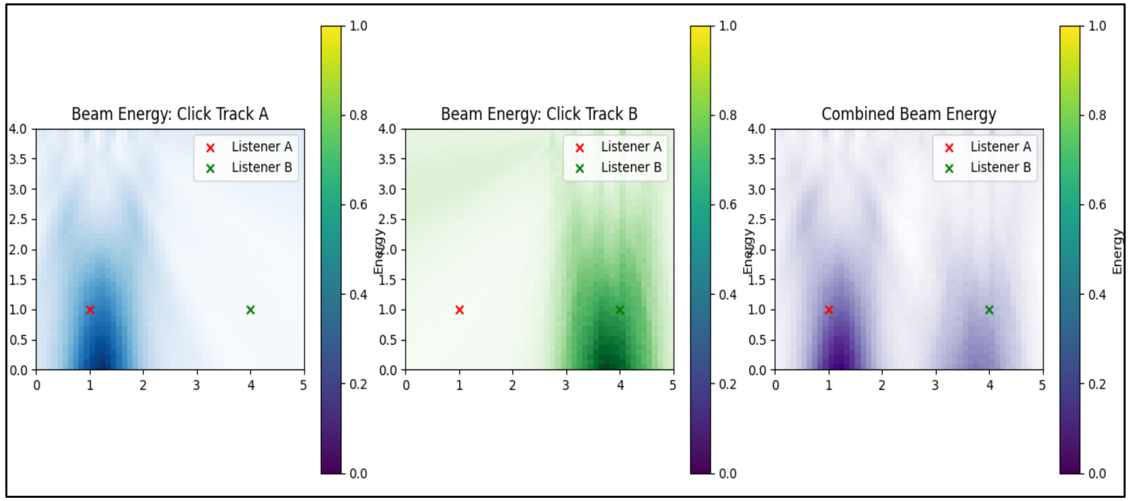
•Different listeners can hear independent audio streams

•Allows for applications like cars, shared rooms, or films in multiple languages

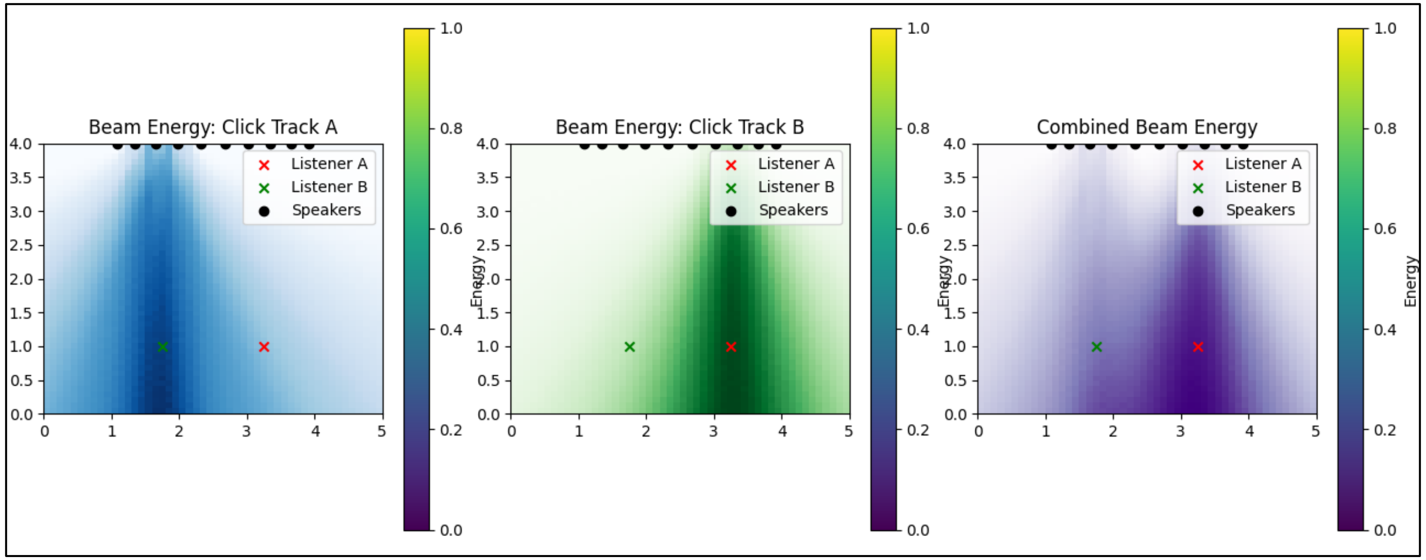
Progress to Date

- Reviewed existing work such as papers, articles, demonstrations on directional audio and beamforming
- Evaluated simulation tools and selected Python due to available DSP and room acoustics libraries
- Developed an initial Python simulation to model speaker positions, listener locations and room geometry
- Implemented listener-specific audio outputs to verify sound separation at each position
- Started with simple click tracks to verify beamforming behaviour and to observe basic audio separation
- Extended to full song snippets to test system performance with different genres and energy mixes
- Established a baseline system using distance-based delay beamforming
- Improved directionality by evaluating array geometry
- Added null steering
- Structured the system so each simulated speaker corresponds to a physical output channel
- Selected hardware for physical prototype implementation



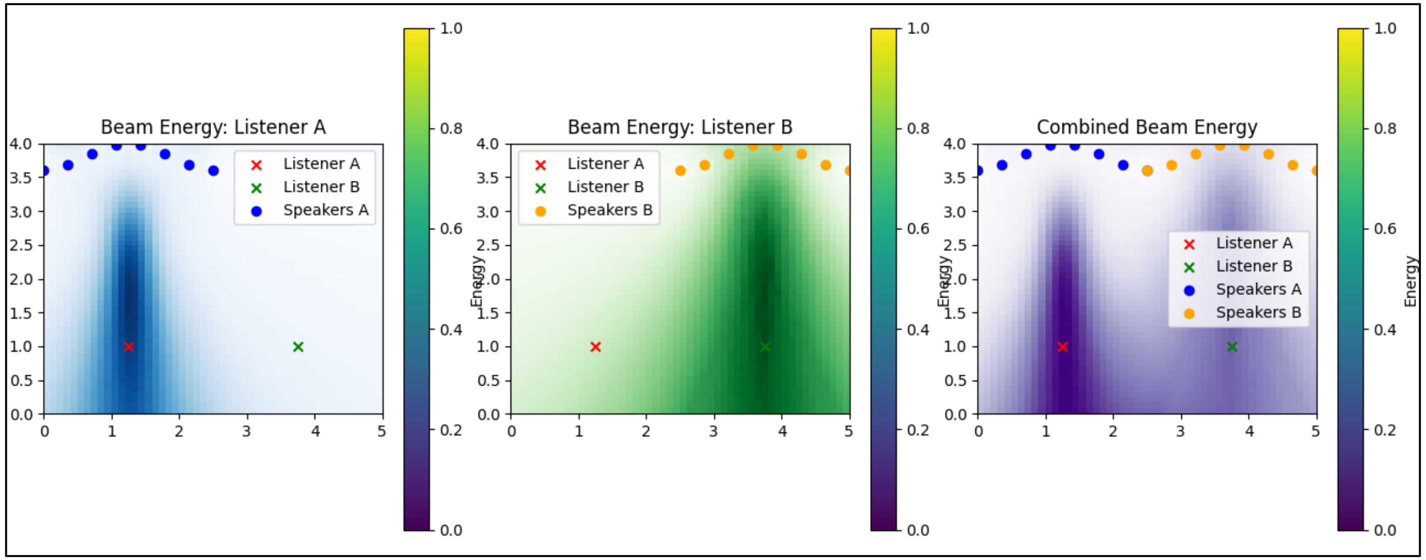


Initial Results



Listener A

Listener B

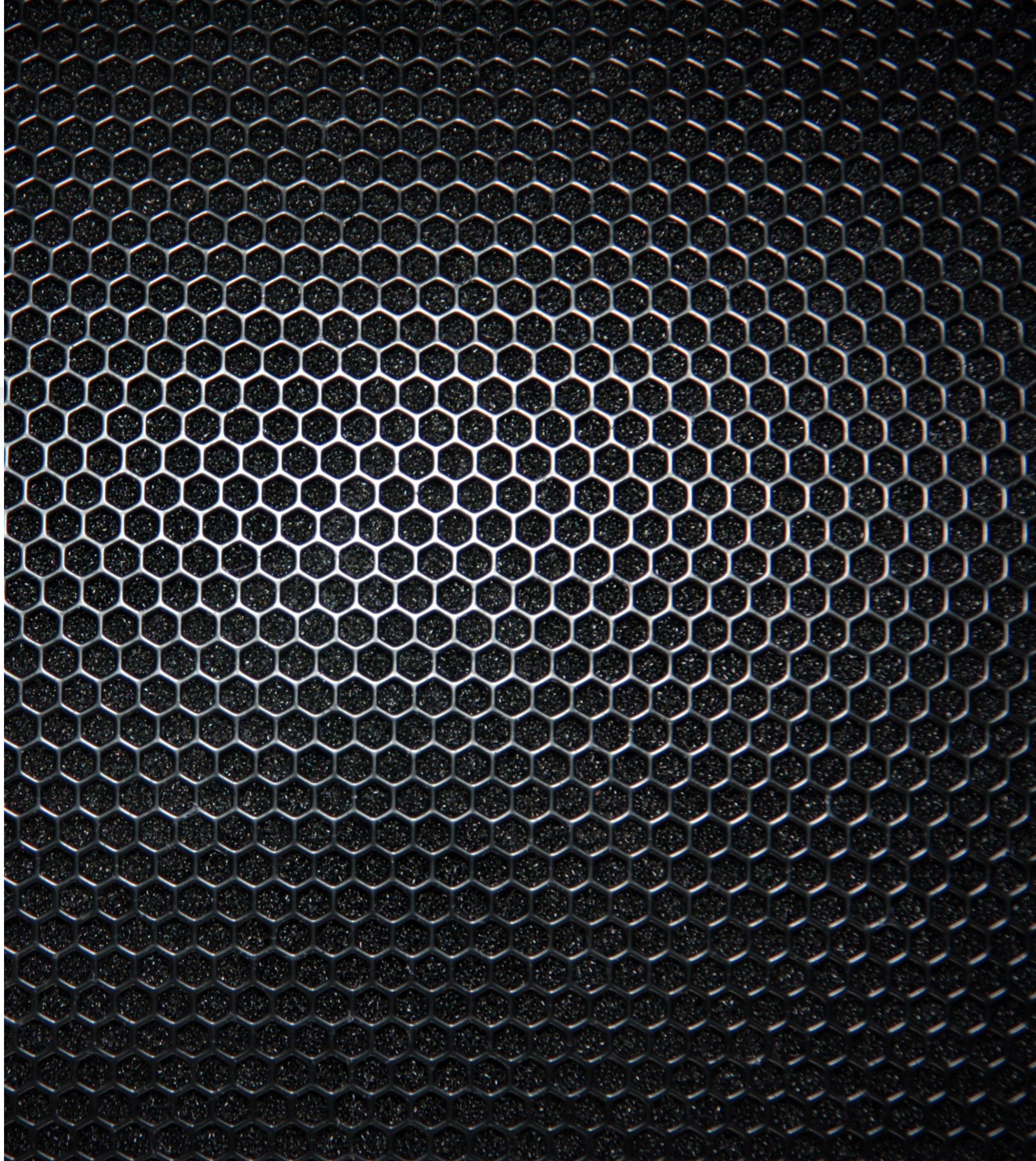


Listener A

Listener B

Initial Results

- Simulation successfully developed and operational
- Beamforming directs sound to target listener positions
- Null steering reduces bleed between audios
- Curved and adjustable speaker arrays tested for better focus
- Each listener position produces a separate audio output
- Energy maps visualise beam patterns and zone separation
- System allows changes to room geometry, speaker array and listener positions



Future Work

- Build the physical speaker array with selected hardware
- Configure arrays to deliver clear, separate audio streams to listeners
- Evaluate and measure beam accuracy, zone separation and bleed of the physical prototype
- Adjust array geometry, delays and null steering for improved directional audio
- Prepare and finalise all project documentation, simulations and demonstration materials

WP #	Task #	Sept	Oct	Nov	Dec	Jan	Feb	Mar	April
WP1	T1	X	XXX						
	T2		XX	XX					
	T3			XX					
WP2	T1		XX	XXXX					
	T2		XX	XXXX					
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WP3	T1					XX			
	T2					X	XXX		
	T3					X	XXX		
WP4	T1						XX	XX	
	T2						XX	XX	
	T3						XX	XX	
WP5	T1							XX	XX
	T2							XX	XX

Conclusion

- The project investigates directional audio using loudspeaker arrays.
- The aim is to provide separate audio to different listeners in the same space with minor bleed between sources.
- Beamforming techniques have been explored through research and put into practice in simulation.
- Additions to the simulation, such as array geometry changes and null steering, improved beam focus and reduced audio bleed.
- Simulation results provide a foundation for building the physical system.
- The remaining work focuses on building, testing and finalising the system.